

Al-Balqa Applied University
Faculty of Engineering Technology
Electrical Engineering Department

Electrical Machines I

1st semester 2009/2010

Final Exam.

- Q1 : Draw the 3-phase circuit of the given single line network diagram (fig.1) and calculate:
1. The voltage of the load. 2. The load current. 3. The power factor. 4. The input current.
5. The capacity of the transformers. (Neglect the magnetizing currents).

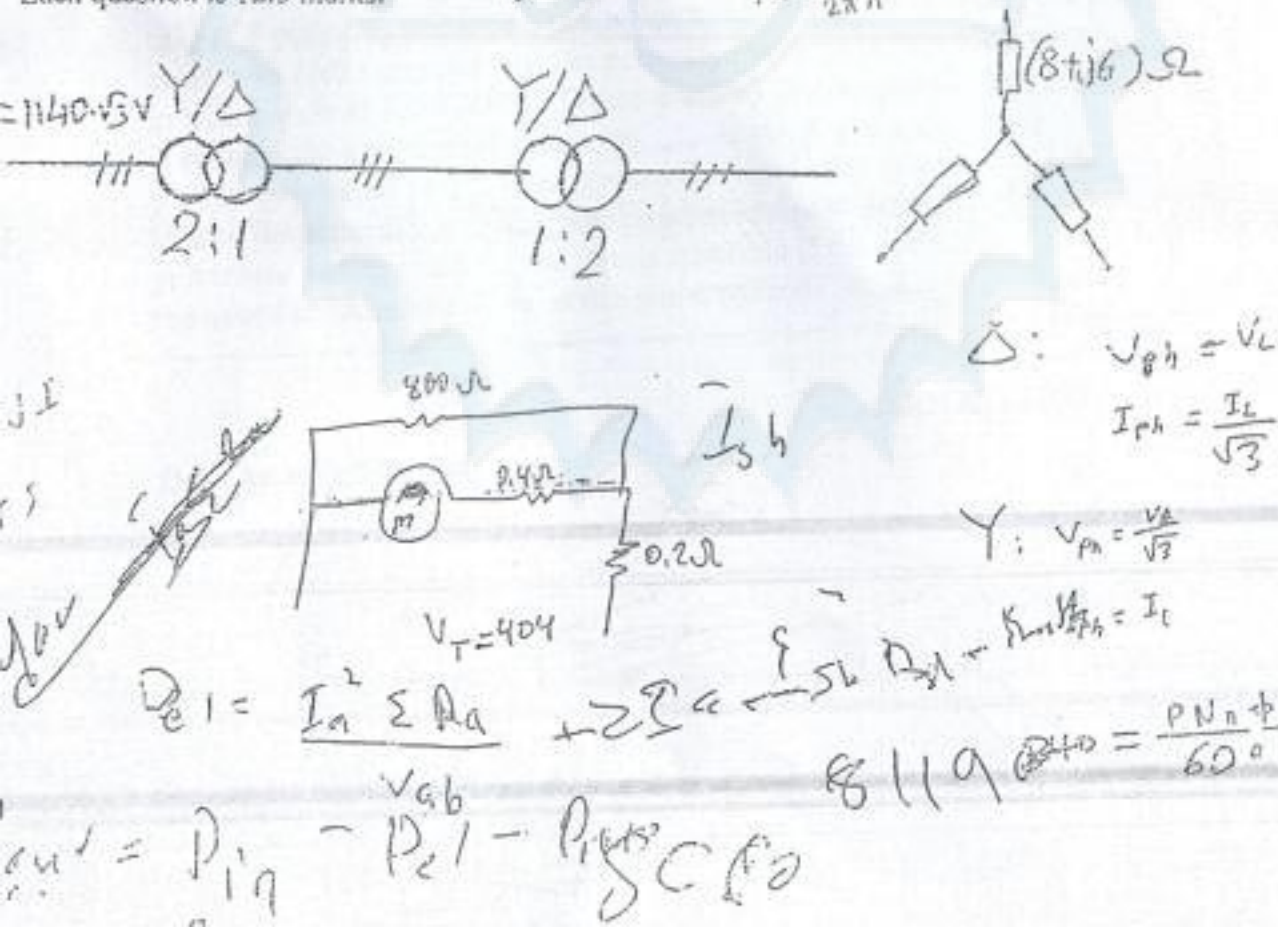
- Q2 : A power transformer has rated capacity 22.8 KVA . The efficiency was maximum and equal 95% at pure active load while the electrical losses were 480 W . Find :
1. The constant losses 2. The rated electrical losses. 3. The efficiency at full load with $\text{PF} = 0.8$.

- Q3 : A- The EMF of DC machine with wave winding is 240 V , the speed 1500 RPM and the current passing each conductor of the armature coil 7.85 A . Calculate the electromagnetic torque.

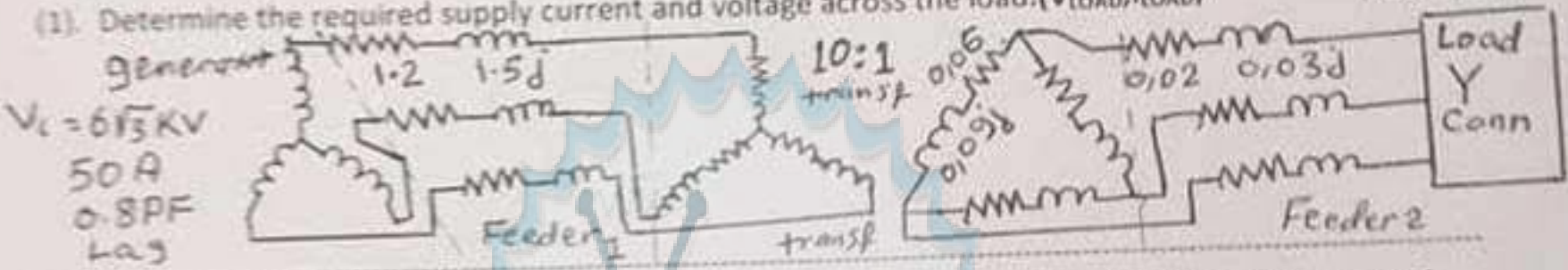
B- Two power transformers have $V_{\text{reg}} = 6\%$ each, one of them is 4^{th} connection group, the other is 8^{th} group. If by fault they were connected in parallel find the circulating current in per unit.

- Q4 : DC compound motor, short shunt, 404 V . The resistances of : series winding is 0.2Ω , shunt winding is 800Ω , armature winding is 0.4Ω , the speed is 1500 RPM , and the rotational losses 266.9 W . Neglecting stray losses find :
1. The output power. 2. The efficiency. 3. The torque on the shaft. 4. The EMF.

Each question is 12.5 marks.



Q1) A 3-phase, $6\sqrt{3}$ KV, 50A at 0.8 power factor lagging, Y-connected generator (power source) supplies a 3-phase, Y-connected load, through a 10:1 step-down, Y/ Δ , transformer, the impedance of the transformer referred to the primary side is $(0.06 + 0.09j)\Omega$. The generator connected with transformer through a feeder whose impedance $(1.2 + 1.5j)\Omega$ per phase. The load connected with transformer through a feeder whose impedance $(0.02 + 0.03j)\Omega$ per phase. The load consumed at power source as shown in figure (1). Determine the required supply current and voltage across the load. (V_{LOAD}, I_{LOAD}) (10-marks)



Q2) A 40KVA, 0.4/6 KV; single phase transformer has an impedance referred to the primary side of $0.12 + 0.16j$. The maximum efficiency of the transformer occurs when primary load current flow in the transformer equal 90 A. Determine:- a) The electrical losses and magnetic losses. (3-marks)
b) The maximum efficiency at 0.9 power factor lag and the efficiency of the transformer when it delivers 24 Kw at 0.8 power factor and find the loading factor. (4-marks)

Q3) A 24.5 Kw, 245V, 1800RPM, shunt DC generator has the following parameters:- $R_A = 0.2\Omega$; $R_{sh} = 70\Omega$; $R_{ad} = 0.105\Omega$; $N_{sh} = 2500$ turns per pole, $F_{ad} = 500$ A turns the prime mover torque $T_{in} = 8\text{N.m}$, and the magnetization curve is shown in fig (2). If $R_{ad} = 28\Omega$ Find:-

- The maximum and the minimum generated voltage at no-load condition (3-marks)
- If $R_{ad} = 28\Omega$; Find:- the generated and terminal voltages (E_A, V_T) at no-load and at full load with out armature reaction and with armature reaction (5-marks)
- If the terminal voltages (V_T) was found to be 220 V What would be the equal armature current equal.
- If the speed of the generator decreased to the 1500 RPM and the terminal voltage is 200V, how much field current be flowing in the generator (2-marks)
- find the maximum load current (I_{max}) the critical speed (n_c) and the critical resistance (R_c) and the steady state short circuit current (4-marks)

Q4) A 25 Kw, 250V, 1800RPM, Separately DC motor has the following parameters:- $R_A = 0.2\Omega$; $R_{sh} = 50\Omega$; $R_{ad} = 150\Omega$. At no-load condition the motor the armature takes current 5 A. If $R_{ad} = 75\Omega$; At 0.8 full load Find:-

- The speed of the motor. (2-marks)
- The electromagnetic torque and the efficiency at 0.8 full load. (4-marks)
- What is the speed of the motor If $R_{ad} = 10\Omega$; and If $R_{ad} = 100\Omega$ with constant (E_A). (2-marks)
- what is the starting current and starting torque (3-marks)
- Find the speed of the motor if the terminal voltage decreased to 200V at 0.8 full load. (2-marks)
- At no-load what happened with motor speed If the field coil suddenly disconnected at source (2-marks)

Q5) a) what is the armature reaction and the major problems caused by Armature reaction (2-marks)
b) In one figure at the same no-load terminal voltage draw the various of the load (V_T) characteristics of the generators and briefly explain load (terminal voltage) characteristics of the shunt dc generator. (4-marks)
b) Derive the mechanical and electromechanical characteristic and draw it for the series DC motor. (2-marks)